PAPER_715: UQFF Knowledge Base 17: THz Q-Scope Signal Analysis (Set 40: Signals 31-40)

Class: `UQFFKnowledgeBaseKB17` **CP4 Entry:** #299 **Keywords:** THz, q-scope, signals 31-40, Ug1 thread, U_bi buoyancy, flow state, UQFF KB17 **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

UQFF Knowledge Base 17 (KB17) presents the detailed analysis of THz q-scope Set 40 (Signals 31-40), captured at 16:46:13–16:48:10 UTC-4 on the Star-Magic q-scope apparatus. Set 40 introduces the UQFF Ug1 thread strength metric and the U_bi buoyancy adjustment as the primary physical analysis framework for THz signal bundles.

1. Instrument Configuration (Set 40)

Parameter	Value
Time/Div	200 ns
Volt/Div	500 mV
Duration	117 s (16:46:13–16:48:10)
Signals	31–40 (10 total)

2. Signal Data

Signal	Ch1 (V)	Ch2 (V)	Flow State
31	0.60	0.35	Normal
32	0.65	0.40	Normal
33	0.60	0.35	Normal
34	0.55	0.30	Chaotic
35	0.50	0.35	Inverted
36	0.60	0.40	Inverted
37	0.55	0.35	Chaotic
38	0.50	0.30	Chaotic
39	0.50	0.35	Inverted
40	0.50	0.35	Inverted

3. Ug1 Thread Strength

$$U_{g1,i} = \mu_J \, \omega_{THz} \, V_{ch1,i}^2$$

Total thread strength across Set 40:

$$\mathcal{T}_{Ug1} = \sum_{i=1}^{10} U_{g1,i} = \mu_J \, \omega_{THz} \sum_i V_i^2$$

4. U_bi Buoyancy Adjustment

$$U_{bi,i}(t) = \rho_{UA} \, \omega_{THz} \, V_{ch1,i} \, \cos\left(\frac{2\pi t}{\tau_{flow}}\right)$$

The buoyancy adjustment modulates as the Earth's core ACE/DCE pressure cycle causes flow reversals in the q-scope magnetic coupling.

5. Combined UQFF Observable

$$[U_m : SM_m \,/\, Ug1 = UQFF_g + SM_g]^{SCm} = \frac{U_{m,bundle}}{Ug1_{thread}} \cdot \frac{\rho_{SCm}}{\rho_{UA}}$$

References

- Q-Scope Laboratory Records Set 40, Star-Magic 2026
- UQFF Framework v5.32, Star-Magic Session 175

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